

cold segments C1..C20: one fresh pinned process per arm on new point s$(i-1)$ (role rotkey-cold-v1)

cP	**cK**	rT
rank probe (baseline)	candidate cold	tuned rho

rank blocks K1..K4 (new seeds 501..504): same 600 attempts in every arm; table must equal B's

kP	**kK**	kKaa	kKX
frame loop	rotation key	A/A	planted: lift_scan redone

Primary (frozen):

$$R = \frac{\overline{\mathrm{rank}}(kP) + \overline{\mathrm{cold}}(cP)}{\overline{\mathrm{rank}}(kK) + \overline{\mathrm{cold}}(cK)} \quad \text{(end-to-end cold)}$$

improvement iff 95% lower bound > 1; bootstrap: blocks, then attempts paired within a block, and points.
Gates: per block one relation digest, one per-attempt probe vector, one root-call count, one set of point-addition counters across kP, kK, kKaa, kKX; 0 rotation-key collisions on the 29,878,182 stored keys.